

MTH:2310 DISCRETE MATH
THE LAST QUIZ: QUIZ 10
DUE: FRIDAY MAY 9

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 2 points for arithmetic errors.

You will be penalized up to 2 points for disorganized or hard to read solutions.

Problem 1. (12 points) *Relations and Their Properties*

- (a) (8 points) Let $R = \{(a, b), (a, c), (b, b), (c, b), (b, c)\}$ be a relation on the set $A = \{a, b, c\}$. Determine whether R is reflexive and/or symmetric and/or antisymmetric and/or transitive. Justify your answer.

not reflexive, missing (c, c) .

not symmetric, $(a, c) \in R$ but $(c, a) \notin R$

not antisymmetric, $(c, b) \in R$ and $(b, c) \in R$ but $c \neq b$

not transitive, $(c, b) \in R$ and $(b, c) \in R$ but $(c, c) \notin R$.

- (b) (4 points) Define a relation R' on the same set A using roster notation such that R' has at least 2 elements and is symmetric and transitive, but not reflexive.

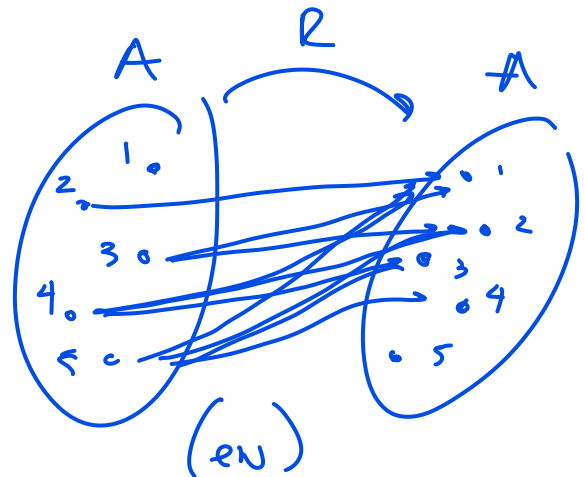
For example $R' = \{(a, a), (b, b)\}$
 $\{(a, a), (a, b), (b, a), (b, b)\}$

Problem 2. (14 points) *Relations and Their Properties*

- (a) (6 points) Let $A = \{1, 2, 3, 4, 5\}$. Let R be the relation where $(x, y) \in R$ if and only if $x > y$. Describe R completely using roster notation, then illustrate the relation as a function or with a table.

$$R = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4)\}$$

R	1	2	3	4	5
1					
2	✓				
3	✓	✓			
4	✓	✓	✓		
5	✓	✓	✓	✓	



- (b) (8 points) Determine whether this relation on A is reflexive and/or symmetric and/or antisymmetric and/or transitive. Justify your answer(s).

not reflexive. No $a \in A$ satisfies $a > a$.

not symmetric. $2 > 1$ but $1 \not> 2$

✓ anti-symmetric. Never have $(a > b)$ and $(b > a)$
so this is "vacuously true"

✓ transitive: If $a, b, c \in A$ and $a > b$, $b > c$
then $a > b > c \Rightarrow a > c$.

Problem 3. (8 points) *Equivalence Relations*

Let R be the relation on all sets so that for any two sets A and B , A is related to B if and only if $A \cap B \neq \emptyset$. Determine whether this is an equivalence relation by checking all relation properties.

Reflexive? What if $A = \emptyset$? Then $\emptyset \cap \emptyset = \emptyset$.

Not reflexive.

Symmetric? Yes. $A \cap B \neq \emptyset$ means they have at least one element in common. Order does not change this.

Anti-symmetric? NO. $A = \{1, 2\}$, $B = \{2, 3\}$. $A \cap B = \{2\} = B \cap A$ but $A \neq B$.

Transitive? NO. $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{3, 4\}$
 $A \cap B = \{2\}$, $B \cap C = \{3\}$ but $A \cap C = \emptyset$.

Problem 4. (12 points) *Equivalence Relations*

Let R be the relation on the set of all bit strings defined so that for any bit strings x and y , xRy if and only if x and y contain the same number of 1's.

(a) (8 points) Show that R is an equivalence relation.

Reflexive: If x is a bit string, it always has as many 1's as itself. $(x, x) \in R$ for all x .

Symmetric: $(x, y) \in R$ means x and y are bit strings with same number of 1's. Order doesn't change that. $(y, x) \in R$.

Transitive: If $(x, y) \in R$ and $(y, z) \in R$ then x, y, z are all bit strings. x and y have same number of 1's and y and z have same number of 1's so x and z do too. $(x, z) \in R$.

(b) (4 points) Describe in words the equivalence class $[10101]_R$, then list 5 of its members.

$[10101]_R$ all bit strings with exactly three 1's.

$[10101]_R = \{ 111, 1110, 1011, 1010100000, 10001000001, \dots \}$

Problem 5. (24 points) *Graphs and Graph Terminology*
Consider the following sets.

$$A = \{1\}$$

$$C = \{3\}$$

$$E = \{2, \{1, 3\}\}$$

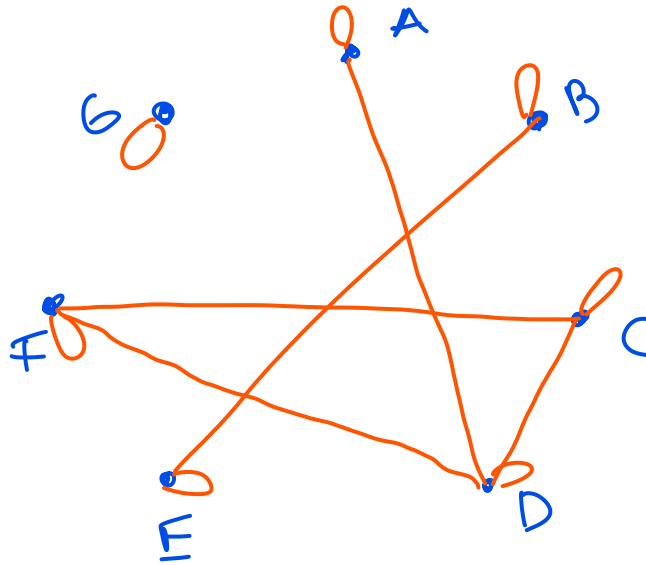
$$G = \{\{2, 3\}, 4\}$$

$$B = \{2\}$$

$$D = \{1, 3\}$$

$$F = \{\{1, 2\}, 3\}$$

- (a) (8 points) Draw a graph where each node represents one of the sets A, B, C, D, E, F , or G . Draw an edge to connect nodes whose corresponding sets have a non-empty intersection. Label your nodes appropriately



$$\begin{aligned} A \cap A &\neq \emptyset \\ B \cap B &\neq \emptyset \\ &\vdots \end{aligned}$$

- (b) (3 points) What kind of graph is this (simple, multi, pseudo, directed, mixed, undirected)?

undirected pseudograph

- (c) (4 points) Find the degree of each vertex.

$$\deg(A) = 3$$

$$\deg(D) = 5$$

$$\deg(B) = 3$$

$$\deg(E) = 3$$

$$\deg(C) = 4$$

$$\deg(F) = 4$$

$$\deg(G) = 2$$

(d) (4 points) Verify the Handshaking Theorem using this graph.

$$2(\# \text{ edges}) = \deg(A) + \deg(B) + \dots + \deg(G)$$

$$2(12) = 3 + 3 + 4 + 5 + 3 + 4 + 2 = 24$$

$$24 = 24 \checkmark$$

(e) (2 points) Find the neighborhood of the node corresponding to the set C (find $N(\{C\})$).

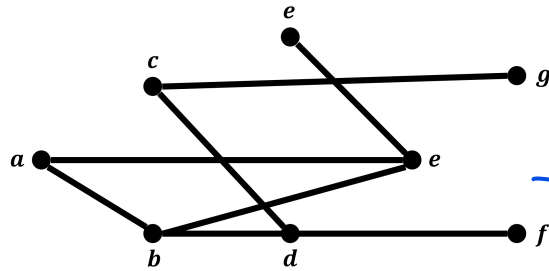
$$N(\{C\}) = N(C) = \{C, F, D\}$$

(f) (3 points) Find the neighborhood of the nodes corresponding to the sets D and G (find $N(\{D, G\})$).

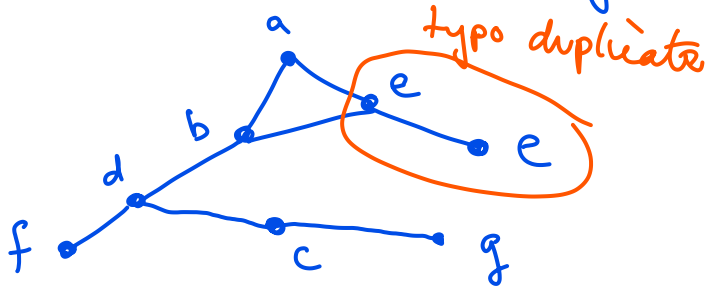
$$\begin{aligned} N(\{D, G\}) &= N(D) \cup N(G) \\ &= \{D, C, A, F\} \cup \{G\} \\ &= \{D, C, A, F, G\} \end{aligned}$$

Problem 6. (4 points) *Introduction to Trees*

Is the following graph a tree? Justify your answer.

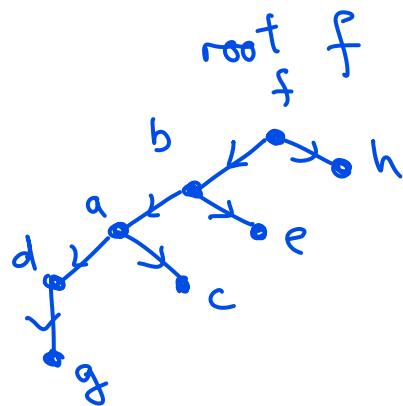
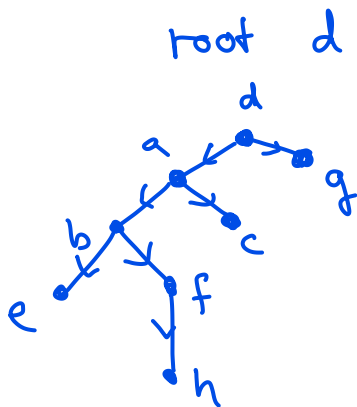
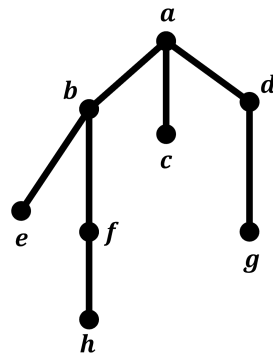


Not a tree. Cycle a, b, e, a



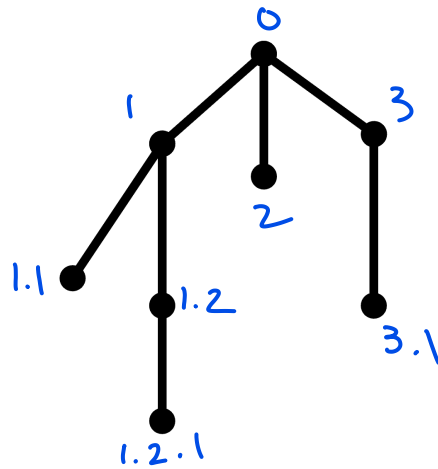
Problem 7. (6 points) *Introduction to Trees*

Find the rooted tree with root d , and the rooted tree with root f using the following tree.



Problem 8. (11 points) *Tree Traversal*

Consider the following unlabeled rooted tree.



- (a) (2 points) Label the tree using the universal address system. **You should label the graph above directly.**
- (b) (3 points) In which order are the vertices of the rooted tree visited using a **pre-order** traversal?

0, 1, 1.1, 1.2, 1.2.1, 2, 3, 3.1

- (c) (3 points) In which order are the vertices of the rooted tree visited using an **in-order** traversal?

1.1, 1, 1.2.1, 1.2, 0, 2, 3.1, 3

- (d) (3 points) In which order are the vertices of the rooted tree visited using a **post-order** traversal?

1.1, 1.2.1, 1.2, 1, 2, 3.1, 3, 0